LLM Content Moderation and User Satisfaction: Evidence from Response Refusals in Chatbot Arena

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Abstract

LLM safety and ethical alignment are widely discussed, but the impact of content moderation on user satisfaction remains underexplored. In particular, little is known about how users respond when models refuse to answer a prompt—one of the primary mechanisms used to enforce ethical boundaries in LLMs. We address this gap by analyzing nearly 50,000 model comparisons from Chatbot Arena, a platform where users indicate their preferred LLM response in pairwise matchups, providing a large-scale setting for studying real-world user preferences. Using a novel RoBERTa-based refusal classifier fine-tuned on a hand-labeled dataset, we distinguish between refusals due to ethical concerns and technical limitations. Our results reveal a substantial refusal penalty: ethical refusals yield significantly lower win rates than both technical refusals and standard responses, indicating that users are especially dissatisfied when models decline a task for ethical reasons. However, this penalty is not uniform. Refusals receive more favorable evaluations when the underlying prompt is highly sensitive (e.g., involving illegal content), and when the refusal is phrased in a detailed and contextually aligned manner. These findings underscore a core tension in LLM design: safety-aligned behaviors may conflict with user expectations, calling for more adaptive moderation strategies that account for context and presentation.

Keywords: Large Language Models (LLMs), Content Moderation, User Trust, Ethical AI, Refusal Behavior, Human-AI Interaction

1. Introduction

With the rise of large language models (LLMs) such as ChatGPT, the topics of LLM alignment and safety have garnered significant attention across academia (Xie et al., 2024), industry (OpenAI, 2023, Antropic, 2023), and regulatory bodies (European Data Protection Board, 2024). LLMs, while demonstrating remarkable capabilities, face challenges due to their black-box nature, which makes controlling their outputs inherently difficult. Consequently, these models may generate illegal, unethical, or inappropriate content, raising concerns about how such systems shape access to information and mediate social norms (Weidinger et al., 2021).

One key strategy to address these risks is for LLMs to refuse to answer certain prompts. Refusals now play a central role in content moderation, acting as a safeguard against generating outputs that violate ethical, legal, or platform-specific constraints. A growing body of technical work has focused on improving *when* and *how* models should refuse, including the development of guardrails (Dong et al., 2024), detoxification techniques (Welbl et al., 2021), fine-tuning methods using ethically curated datasets (Raza et al., 2024), and safety benchmarks that evaluate refusal consistency across sensitive prompts (Xie et al., 2024). However, much

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of this research centers on optimizing refusal behavior from a system perspective—while leaving open the question of how such refusals are actually received by users.
Understanding user perceptions of content moderation due to ethical concerns is increasingly important, as it shapes both public trust and the broader acceptance of AI systems in real-world applications. While users generally support the idea of safeguards against harmful or inappropriate content (Kieslich et al., 2021), overly restrictive or poorly phrased refusals may be seen as unhelpful, opaque, or even paternalistic (Eslami et al., 2019). These dynamics can undermine not just individual user experience, but also trust in the system's legitimacy and alignment with social expectations (Ananny & Crawford, 2018). This tension between safety and helpfulness reflects a growing sociotechnical challenge and remains an underexplored dimension in current LLM research.

In fact, refusals triggered by ethical or safety concerns may hurt user satisfaction because they violate expectations for cooperation (Burgoon, 1993), disrupt conversational flow (Koudenburg et al, 2013), or come across as overly restrictive or moralizing (Kieslich et al., 2021). Potentially, refusals due to ethical concerns or content moderation may be perceived even more negatively than those stemming from technical or functional limitations, such as the inability to access real-time data. While users may accept the latter as an unavoidable system constraint, refusals rooted in safety policies may feel less justified or overly cautious, especially when applied without explanation.

Correspondingly, this paper focuses on the following primary research questions:

- **RQ1:** Do LLM response refusals negatively affect user satisfaction?
- **RQ2:** Are refusals due to ethical concerns (i.e., content moderation) perceived more negatively than refusals stemming from technical limitations?

To address this gap, we leverage data from Chatbot Arena, a platform where users compare responses from two LLMs and indicate their preferred option (Chiang et al., 2024). Chatbot Arena provides a natural setting to study user preferences, as it captures revealed preferences in head-to-head matchups between models. However, a technical challenge arises in classifying responses that involve refusals. While existing approaches like SORRY-Bench evaluate compliance on pre-defined prompts related to ethical concerns (Xie et al. 2024), they do not distinguish between different types of refusals. This makes it challenging to identify refusals in Chatbot Arena answers due to content moderation on sensitive issues from other reasons for response refusals, such as technical inabilities of the corresponding model.

To overcome this limitation, we fine-tune a RoBERTa-Large model on a hand-labeled dataset of responses, enabling the model to classify refusal behaviors based on their underlying reason: (i) ethical concerns, or (ii) lack of technical capabilities or insufficient context or information. Additionally, we distinguish between responses that completely refuse to conduct the task and those that provide disclaimers but attempt to address the task of the prompt.

Using this classifier, we analyze almost 50,000 one-turn prompts and response pairs from Chatbot Arena, with a particular focus on refusals to conduct tasks due to ethical concerns. Our analysis reveals a significant "refusal penalty": when models refuse to answer due to ethical concerns, their win rate drops to 8%, compared to 36% for normal responses. The penalty becomes even more pronounced when refusals are paired against normal responses, with win rates as low as 4% for ethical-based refusals. This penalty is notably stronger than for technical refusals, which also reduce win rates but to a lesser extent (16% win rate overall; 13% when paired against normal responses).

Moreover, to better understand and potentially mitigate this refusal penalty, we explore two further dimensions. First, we investigate how the phrasing and presentation of refusals affect user preferences—focusing on features like response length and textual similarity to the user prompt. Second, we examine whether refusals are judged more favorably when they respond to prompts that are clearly inappropriate or harmful, as identified by a content moderation API.

Taken together, beyond the core link between refusal behavior and user satisfaction, this study addresses two extension questions:

- **RQ3:** Does the phrasing of a refusal moderate its impact on user satisfaction?
- **RQ4:** Are refusals judged more favorably when triggered by clearly sensitive or unsafe content?

In summary, this paper makes three core contributions. First, we develop a novel typology of LLM refusal behavior, distinguishing between ethical and technical refusals, as well as disclaimers, and operationalize this framework through a fine-tuned classifier. To support future research, we publicly release our refusal classifier and annotated dataset.[2] Second, we provide large-scale empirical evidence of a refusal penalty in user evaluations—especially for ethical refusals—using real-world data from Chatbot Arena. Third, we identify key moderators of this effect, showing how both phrasing (response length and alignment) and prompt sensitivity (moderation category) shape user and model-based perceptions of refusal behavior. Together, these findings offer new insights into the tradeoffs between alignment, helpfulness, and user satisfaction—highlighting design implications for more effective and context-aware moderation in LLM systems.

2. Theoretical Background

2.1. LLM Response Refusals and User Satisfaction

As large language models (LLMs) become widely integrated into everyday applications, users increasingly expect them to be capable, responsive, and goal-oriented. These expectations are not only shaped by the technical capacities of the models but also by the growing cultural perception of LLMs as intelligent assistants. Refusal responses—where the model declines to

[2] https://huggingface.co/Human-CentricAI/LLM-Refusal-Classifier
https://huggingface.co/datasets/Human-CentricAI/llm-refusal-chatbot-arena